Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Multiple equivalent characterizations of the Schrödinger equation

Files.

- File: `briefs/SchroedingerEvolution/SchroedingerEvolution.lean`
- Library: `SchroedingerEvolution`
- Imports:
  `ForPhyslib.QuantumMechanics.Wigner.OneParameter` and the brief `UnitaryEntropy` (for `born`, `vonNeumann` and `pure_and_born_of_preserves`)
- Wigner library, new files in `ForPhyslib/QuantumMechanics/Wigner/`: `Kadison` (mixed-state extension, the two-state bound, Parseval averaging, Kadison's theorem without continuity or injectivity, pure states as extreme points, the three-way equivalence), `OneParameter` (exclusion of anti-unitaries by composition, the phase-free conjugation group on $\mathbb{R}$, its generator as an inner $*$-derivation, the lift to a unitary one-parameter group). `Density` gained finite convex combinations, affine maps of density operators, conjugation `conjOp` by a unitary or an anti-unitary, and the spectral decomposition. `OneParameter` imports physlib's

`Physlib.Mathematics.ForMathlib.OneParameterSubgroups.Unitary` (bounded Stone theorem, `UnitaryOneParameterGroup`).

- New file `ForPhyslib/QuantumMechanics/Wigner/EqualMixtures.lean`: a map that preserves equal mixtures of density operators is affine on them; `IsDensity.norm_le_one` moved here from `VonNeumannResolvent`.

- New `ForMathlib` material:
  `ForMathlib/Analysis/Convex/Midpoint.lean` (a bounded solution of Jensen's functional equation on $[0,1]$ is affine, so a bounded map preserving midpoints on a convex set preserves convex combinations),
  `ForMathlib/Analysis/InnerProductSpace/Positive.lean` (a positive operator kills every vector on which its quadratic form vanishes),
  `ForMathlib/Analysis/Calculus/Deriv/ContinuousLinearMapApply.lean` (derivative of $s \mapsto f(s)\,x$ for operators linear over a larger field),
  and in `ForMathlib/Analysis/InnerProductSpace/LinearMap.lean`: a norm-preserving ring endomorphism of $\mathbb{C}$ is the identity or conjugation, semilinear isometries conjugate the inner product, and every continuous derivation of the bounded operators on an inner product space is inner (`InnerProductSpace.exists_eq_mul_sub_mul_of_map_mul`).

Setting. E is a finite-dimensional complex Hilbert space, with density operators, pure states `proj` $\psi = |\psi\rangle\langle\psi|$, `born` and `vonNeumann` as in the brief "Schrödinger equation is equivalent to information conservation". A map T on operators is `AffineOnDensities` if $T(t\rho + (1-t)\sigma) = tT\rho + (1-t)T\sigma$ for density operators ρ, σ and $t \in [0,1]$. `IsPure` ρ means $\rho = P_\psi$ for a unit ψ, `BijOnPure` T that T maps the pure states bijectively onto the pure states, and `ImplementedByUnitaryOrAntiunitary` T that $T\rho = W\rho W^{-1}$ on density operators for a unitary or an anti-unitary W. An evolution is a family Φ_t of maps on operators, used for $t \geq 0$ and on density operators only. `IsEvolution` Φ (QST:H + [B]EVMIX + EVTI) says that each Φ_t maps density operators to density operators and is affine on them, $\Phi_0 = \mathrm{id}$, $\Phi_{s+t} = \Phi_s \circ \Phi_t$, and $t \mapsto \Phi_t(\rho)$ is continuous at 0^+. The conditions are

- `DrSceq` $\hbar$: $\Phi_t(\rho) = U_t \rho U_t^\dagger$ with $U_0 = 1$ and $U'(t) = (i\hbar)^{-1} H U(t)$ for a self-adjoint H;

- `DrUnit`: $\Phi_t(\rho) = U_t \rho U_t^\dagger$ for physlib's `UnitaryOneParameterGroup` U (unitary, $U_0 = 1$, $U_{s+t} = U_s U_t$ on $\mathbb{R}$, norm-continuous);

- `DrEv`: each Φ_t is `BijOnPure`;

- `DrProb`: each Φ_t maps pure states to pure states (`MapsPure`) and preserves the Born rule between them (`PreservesBorn`);

- `DrInfo`: each Φ_t preserves the von Neumann entropy of density operators.

Results.

Brief	Lean
Proposition 1, complete	`affine_pure_state_correspondence` (from `Wigner.kadison_tfae`)
Proposition 1, (i) ⇒ (iii)	`Wigner.kadison`: onto the pure states suffices
Proposition 1, mixed states from pure states	`Wigner.AffineOnDensities.eq_conjOp`
Proposition 1, (iii) ⇒ (ii)	`Wigner.bijOn_of_eq_conjOp`
Proposition 1, extreme points are the pure states	`Wigner.isPure_iff_isExtremeDensity`
Proposition 1, (ii) ⇒ (i)	`Wigner.AffineOnDensities.bijOnPure`
Remark after Proposition 1, injective suffices	`bijOnPure_of_injOn` (from `Wigner.AffineOnDensities.bijOnPure_of_injOn`)
Remark, orthogonal pure states stay orthogonal	`Wigner.AffineOnDensities.inner_eq_zero_of_injOn`
Proposition 1 for maps preserving equal mixtures	`affine_pure_state_correspondence_of_map_mix` (from `Wigner.affineOnDensities_of_map_mix`)
Equal mixtures preserved ⇒ affine	`Wigner.affineOnDensities_of_map_mix`, from `Convex.map_add_of_map_midpoint` and `eq_sub_smul_add_smul_of_map_midpoint`
Theorem 3, DR-EV ⇒ DR-UNIT	`DrEv.drUnit`
Theorem 3, anti-unitary case excluded by $\Phi_t = \Phi_{t/2}^2$	`Wigner.exists_unitary_of_map_add`
Theorem 3, "the phases can be chosen"	`Wigner.exists_addChar_adL`, `Wigner.exists_unitaryOneParameterGroup_of_addChar`, `Wigner.exists_unitaryOneParameterGroup`, `drUnit_of_implemented`
Theorem 3, DR-PROB ⇒ DR-UNIT	`DrProb.drUnit` (via `implemented_of_mapsPure`)
Theorem 3, DR-INFO ⇒ DR-UNIT	`DrInfo.drProb`, then `DrProb.drUnit`
Theorem 3, DR-UNIT ⇒ DR-EV, DR-PROB, DR-INFO	`DrUnit.drEv`, `DrUnit.drProb`, `DrUnit.drInfo`
Theorem 3, DR-UNIT ⇔ DR-SCEQ	`drUnit_iff_drSceq` (for $\hbar \neq 0$), `eq_exp_of_schrodinger`
Theorem 3, complete	`schrodinger_tfae` (for $\hbar \neq 0$)

Differences from the brief.

- Finite dimension only; the brief has a separable Hilbert space. In finite dimension the trace class is all of $\mathrm{End}(E)$, strong and norm continuity agree, and the generator is bounded, so Stone's theorem is physlib's bounded version.
- The brief's evolution is defined for $t \geq 0$ only ([B]EVMIX has $t_1 \geq t_0$), so all conditions quantify over $t \geq 0$. DR-UNIT still asserts a unitary group on all of $\mathbb{R}$, as the brief's $U_{-t} = U_t^\dagger$ does.
- Joint continuity of $(\rho, t_0, t_1) \mapsto \Phi_{t_0 \to t_1}(\rho)$ is weakened to continuity of $t \mapsto \Phi_t(\rho)$ at 0^+ for each density operator ρ, which joint continuity implies. Proposition 1 needs no continuity at all. The two-parameter form with joint continuity is in Section 1.1.

- DR-SCEQ is stated with a constant H; the brief writes $H(t)$. The brief's form is in Section 1.1, where it is shown to be equivalent under EVTI. It is stated for the propagator U, which gives the brief's form $\psi(t) = U(t)\psi(0)$ for every initial vector.

- The "reconstructed past" clauses of DR-PROB and DR-INFO are dropped: a past state ψ_P is one with $\Phi_t(\psi_P) = \psi_F$, so its equality follows from the future one. In DR-EV they become injectivity and surjectivity on pure states.

- DR-PROB $\Rightarrow$ DR-UNIT goes directly through Wigner's theorem in its non-bijective form (Gehér) and the mixed-state extension, not through DR-EV. DR-INFO $\Rightarrow$ DR-UNIT goes through DR-PROB: zero entropy characterizes pure states, and the entropy of an equal mixture of two pure states determines their transition probability (`pure_and_born_of_preserves`, with equal mixtures preserved by affinity, `Wigner.AffineOnDensities.map_mix`). Neither needs bijectivity on pure states.

- DR-UNIT $\Rightarrow$ DR-PROB compares $\mathrm{tr}(P_{\psi'}P_{\phi'}) = |\langle\psi'|\phi'\rangle|^2$, so the phases of the image vectors never enter.

- The proof of DR-SCEQ $\Rightarrow$ DR-UNIT shows $U_t = \exp(t(i\hbar)^{-1}H)$ because $\exp(-t(i\hbar)^{-1}H)U_t$ has derivative zero; it needs no $\hbar \neq 0$. The converse, $H = \hbar\cdot$generator, does.

- `DrSceq` and `DrUnit` use the adjoint and physlib's `UnitaryOneParameterGroup`, which require `CompleteSpaceE`. The brief file therefore carries `[CompleteSpaceE]` next to `[FiniteDimensional ℂ E]`, although the second implies the first.

Problems found in the brief. No errors in the statements. Gaps and implicit points:

- Proposition 1, (i) $\Rightarrow$ (iii), rests on "the standard linear rank-one preserver result" without a reference; it is not in any Lean library. In finite dimension it is replaced by a direct argument: the two-state decomposition $\frac{1}{2}(P_\psi + P_\phi) = \lambda P_{\hat{r}} + (1-\lambda)P_{\hat{s}}$ with $\lambda = (1 + |\langle\psi|\phi\rangle|)/2$ shows that affinity cannot decrease $|\langle\psi|\phi\rangle|$, Parseval averaging over an orthonormal basis turns this into equality, and Wigner's theorem finishes. Only surjectivity onto the pure states is used; injectivity is not needed, and surjectivity cannot be dropped (the constant map $\rho \mapsto P_e$ is affine and maps pure states to pure states).

- The Remark claims that injectivity on the pure states suffices in finite dimension "as the compactness of $P(H)$ will force T to be bijective". The claim is true (for T mapping pure states to pure states), but compactness alone does not give it: $x \mapsto x/2$ is a continuous injective, not surjective self-map of $[0,1]$. Invariance of domain on the manifold $P(\mathbb{C}^n)$ would, but it is not in any Lean library. The formalization (`bijOnPure_of_injOn`) uses no topology. For orthogonal unit ψ, ϕ with images P_a, P_b, let L be the linear extension of T and $N = L(|\phi\rangle\langle\psi|)$. Then $A(z) = L(P_{\psi+z\phi}) = P_a + |z|^2P_b + zN + \bar{z}N^\dagger$ satisfies $A(z)^2 = (1+|z|^2)A(z)$, and comparing coefficients of z gives $P_aN + NP_a = N$, $P_bN + NP_b = N$ and $N^2 = 0$. Such an N is $|Na\rangle\langle a|$ or $|a\rangle\langle N^\dagger a|$, and likewise for b, which forces $N = 0$ unless $a \perp b$ or $|\langle a|b\rangle| = 1$; and $N = 0$ makes $A(1) = P_a + P_b$, which is twice a pure state only if $|\langle a|b\rangle| = 1$. Injectivity excludes $|\langle a|b\rangle| = 1$, so orthogonality is preserved. The images of an orthonormal basis are then orthonormal, Bessel's inequality replaces surjectivity in the Parseval averaging above, and Wigner's theorem finishes.

- "The phases can be chosen so that $U_0 = I$ and $U_{t+s} = U_tU_s$" is a lifting theorem (Bargmann's theorem for $\mathbb{R}$) that the brief does not prove. In finite dimension it is avoided by working

with the phase-free maps $A_t = U_t(\cdot)U_t^\dagger$ on $\mathrm{End}(E)$: they form a norm-continuous group on $\mathbb{R}$ (the pure states span $\mathrm{End}(E)$), so $A_t = \exp(tL)$ in $\mathrm{End}(\mathrm{End}(E))$. L is a $*$-derivation, every derivation of $\mathrm{End}(E)$ is inner, $L = [K, \cdot\,]$, and the skew part $K_0 = \frac{1}{2}(K - K^\dagger)$ has the same commutator. Then $V_t = \exp(tK_0)$ is the required unitary group, and $A_t = V_t(\cdot)V_t^\dagger$.

- The exclusion of anti-unitaries by $\Phi_t = \Phi_{t/2} \circ \Phi_{t/2}$ needs EVTI and the group law. In the brief "Schrödinger equation is equivalent to information conservation" it was not available, and continuity was used instead.
- DR-PROB ⇒ DR-EV: "$p = 1$ iff same state" gives injectivity on pure states; surjectivity has to come from the reconstructed past states. The direct route through Wigner's theorem needs neither.
- DR-INFO ⇒ DR-UNIT: zero entropy shows that pure states, and only pure states, map to pure states. Bijectivity on the pure states, which Proposition 1 needs, does not follow from this alone: injectivity needs the entropy of mixtures, surjectivity the reconstructed past. The route through DR-PROB needs neither, and no continuity is used in this step.
- The overall phase of U_t and the additive constant in H (the brief's last remark) appear in the lift as the freedom in K: $K + c$ has the same commutator for every scalar c. Taking the skew part removes the real part of c; the imaginary part is that constant.

1.1 The conditions as the brief writes them

Files.

- File: `briefs/SchroedingerEvolution/SchroedingerEvolution/TwoParameter.lean`
- Library: `SchroedingerEvolution`
- Imports:
  the finite-dimensional part `SchroedingerEvolution`, which gained `hasDerivAt_unitaryOneParameterGroup` (a unitary group solves the Schrödinger equation with $H = \hbar\cdot$generator), split off from `drUnit_iff_drSceq`

Setting. As in finite dimension above, with the brief's two-parameter evolution in place of the one-parameter Φ_t. A family $\Psi_{t_0t_1} = \Phi_{t_0\to t_1}$ of maps on operators is used for $t_0 \le t_1$ and on density operators only. `TwoParameter.IsEvolution` Ψ (QST:H + [B]EVMIX) says that each $\Psi_{t_0t_1}$ maps density operators to density operators and is affine on them, $\Psi_{tt} = \mathrm{id}$, $\Psi_{t_1t_2} \circ \Psi_{t_0t_1} = \Psi_{t_0t_2}$ for $t_0 \le t_1 \le t_2$, and $(\rho, t_0, t_1) \mapsto \Psi_{t_0t_1}(\rho)$ is jointly continuous on the density operators and $t_0 \le t_1$. `TwoParameter.Evti` $\Psi\ \Phi$ (EVTI) says $\Psi_{t_0t_1} = \Phi_{t_1-t_0}$ on density operators. `TwoParameter.DrSceq` $\hbar\ \Psi$ is DR-SCEQ with a time-dependent Hamiltonian: there are self-adjoint $H(t)$ and U with $U(0) = 1$ and $U'(t) = (i\hbar)^{-1}H(t)U(t)$ for every $t \in \mathbb{R}$, and $\Psi_{t_0t_1}(\rho) = V\rho V^\dagger$ with the propagator $V = U(t_1)U(t_0)^\dagger$. DR-UNIT, DR-EV, DR-PROB and DR-INFO are the conditions of the one-parameter section on Φ, since the brief states them for Φ_t.

Results.

Brief	Lean
[B]EVMIX + EVTI give the one-parameter evolution	`TwoParameter.IsEvolution.isEvolution`
DR-SCEQ, the solution is unitary for any self-adjoint $H(t)$	`TwoParameter.star_mul_self_of_hasDerivAt`
Theorem 3, DR-SCEQ with $H(t) \Leftrightarrow$ DR-SCEQ with constant H	`TwoParameter.drSceq_iff` (for $\hbar \neq 0$)
Theorem 3, as stated	`TwoParameter.schrodinger_tfae` (for $\hbar \neq 0$)

Remarks.

- Joint continuity is used only at $t_0 = t_1 = 0$, along $t \mapsto (\rho, 0, t)$, to get the continuity of $t \mapsto \Phi_t(\rho)$ at 0^+ that `IsEvolution` asks for.
- DR-SCEQ with $H(t) \Rightarrow$ DR-UNIT: $\frac{d}{dt}(U^\dagger U) = U^\dagger(\overline{(i\hbar)^{-1}} + (i\hbar)^{-1})H(t)U = 0$, so $U(t)^\dagger U(t) = U(0)^\dagger U(0) = 1$. This needs no continuity of H, and in finite dimension a left inverse is a right inverse, so every $U(t)$ is unitary. Then $\Phi_t = \Psi_{0t}$ is conjugation by $U(t)$, and `drUnit_of_implemented` gives DR-UNIT.
- The converse takes the constant $H(t) = \hbar\cdot$generator of the DR-UNIT group V; the propagator is $V_{t_1} V_{t_0}^\dagger = V_{t_1 - t_0}$ by the group law, and EVTI gives $\Psi_{t_0 t_1} = \Phi_{t_1 - t_0}$.
- That $H(t)$ is then constant up to a time-dependent real multiple of the identity, $H(t) = H + c(t)$, is not formalized; Theorem 3 does not need it. Under EVTI the propagators equal $e^{(t_1 - t_0)(i\hbar)^{-1}H}$ up to a phase, which is where $c(t)$ comes from (the brief's remark on the phase of the unitary).

1.2 Hilbert spaces of any dimension

Files.

- File: `briefs/SchroedingerEvolution/SchroedingerEvolution/InfiniteDimensional.lean`
- File: `briefs/SchroedingerEvolution/SchroedingerEvolution/ShiftSemigroup.lean` (the shift semigroup, see "Problems found in the brief"), importing `InfiniteDimensional` and `Mathlib.MeasureTheory.Function.LpSpace.DomAct.Continuous` (translation is continuous on L^p)
- Library: `SchroedingerEvolution`, which now builds every module `SchroedingerEvolution.*`
- Imports:
  `ForPhyslib.QuantumMechanics.Stone`,
  `UnitaryEntropy.InfiniteDimensional` (for `pure_and_born_of_preserves_vonNeumannEntropy` and `exists_proj_of_past`),
  and the finite-dimensional part `SchroedingerEvolution` (for `MapsPure`, `PreservesBorn` and `mul_mul_star_eq_conjOp`, none of which assumes finite dimension)

- New files in `ForPhyslib/ProbabilisticTheory/HilbertSpace/TraceClass/`: `Compact` (Hilbert–Schmidt and trace-class operators are compact) and `CanonicalExpansion` (a positive trace-class operator is $\sum_i \mu_i |b_i\rangle \langle b_i|$ in the trace norm, with the tail estimate).
- New file `ForPhyslib/QuantumMechanics/TraceNorm.lean`: the map `AffineOnDensityOperators` and, for later use, the facts that such maps contract the trace norm, that conjugation by a strongly continuous family of contractions is trace-norm continuous, and that the pure states are closed. Only the definition is used here.
- New file `ForPhyslib/QuantumMechanics/Kadison.lean`: Kadison's theorem in any dimension, without continuity.
- New file `ForPhyslib/QuantumMechanics/EqualMixtures.lean`: a map that preserves equal mixtures of density operators is `AffineOnDensityOperators`.
- New file `ForPhyslib/QuantumMechanics/Bargmann.lean`: strongly continuous unitary groups, and Bargmann's theorem for $\mathbb{R}$, the lift of a continuous evolution by unitary or anti-unitary conjugations to a strongly continuous unitary group.
- New file `ForPhyslib/QuantumMechanics/Stone.lean`: Stone's theorem in Schrödinger form, on top of PhyslibAlpha's Stone theorem (`PhyslibAlpha/ProbabilisticTheory/HilbertSpace/Unbounded/Existence/StoneReconstruction`) and unbounded spectral theorem.
- New `ForMathlib` material: `ForMathlib/MeasureTheory/Integral/IntervalIntegral/LocalCocycle.lean` (a continuous local 2-cocycle on an interval of $\mathbb{R}$ is a coboundary; the circle-valued form).

Setting. E is a complex Hilbert space of any dimension; separability is not used. Density operators (`IsDensityOperator`), pure states and the entropy (`vonNeumannEntropy`) are as in the section "Hilbert spaces of any dimension" of `briefs/UnitaryEntropy/formalization-log.tex`. A map T on operators is `AffineOnDensityOperators` if $T(t\rho + (1-t)\sigma) = tT\rho + (1-t)T\sigma$ for density operators ρ, σ and $t \in [0,1]$, and `ImplementedOnDensityOperators` T means $T\rho = W\rho W^{-1}$ on density operators for a unitary or an anti-unitary W. A `StronglyContinuousUnitaryGroup` is a family of unitaries U_t, $t \in \mathbb{R}$, with $U_0 = 1$, $U_{s+t} = U_s U_t$ and $t \mapsto U_t x$ continuous for every x. `SolvesSchrodinger` $\hbar\, H\, U$, for a partially defined operator H and bounded operators U_t, says that for ψ in the domain of H every $U_t\psi$ is in the domain and $\frac{d}{dt}U_t\psi = (i\hbar)^{-1}HU_t\psi$. `IsEvolution` is as in finite dimension, with `IsDensityOperator` and `AffineOnDensityOperators`; continuity of $t \mapsto \Phi_t(\rho)$ at 0^+ is in operator norm. The conditions are

- `DrSceq` $\hbar$: $\Phi_t(\rho) = U_t \rho U_t^\dagger$ with $U_0 = 1$ and `SolvesSchrodinger` $\hbar\, H\, U$ for a self-adjoint H;
- `DrUnit`: $\Phi_t(\rho) = U_t \rho U_t^\dagger$ for a `StronglyContinuousUnitaryGroup` U;
- `DrEv`: each Φ_t is `BijOnPure`;
- `DrProb`: each Φ_t satisfies `MapsPure` and `PreservesBorn`, and every pure state is Φ_t of a pure state (`HasPurePast`);
- `DrInfo`: each Φ_t preserves the von Neumann entropy, and every density operator is Φ_t of a density operator.

Results.

Brief	Lean
Proposition 1, complete	`affine_pure_state_correspondence` (from `AffineOnDensityOperators.tfae`)
Proposition 1 for maps preserving equal mixtures	`affine_pure_state_correspondence_of_map_mix` (from `affineOnDensityOperators_of_map_mix`)
Proposition 1, orthogonal pure states stay orthogonal	`AffineOnDensityOperators.inner_eq_zero`
Proposition 1, overlaps preserved	`AffineOnDensityOperators.norm_inner_eq`
Proposition 1, (i) ⇒ (iii) on pure states	`AffineOnDensityOperators.exists_unitary_or_antiunitary`
Proposition 1, mixed states from pure states	`AffineOnDensityOperators.eq_conjOp`
Proposition 1, conjugation preserves density operators	`IsDensityOperator.conjOp`
Proposition 1, extreme points are the pure states	`isPure_iff_isExtremeDensityOperator`
Theorem 3, DR-UNIT ⇔ DR-SCEQ	`drUnit_iff_drSceq` (for $\hbar \neq 0$), from `StronglyContinuousUnitaryGroup.exists_solvesSchrodinger` and `exists_stronglyContinuousUnitaryGroup_of_solvesSchrodinger`
Theorem 3, DR-UNIT ⇒ DR-EV, DR-PROB, DR-INFO	`DrUnit.drEv`, `DrUnit.drProb`, `DrUnit.drInfo`
Theorem 3, DR-EV ⇒ DR-UNIT	`DrEv.drUnit`
Theorem 3, anti-unitaries excluded and "the phases can be chosen"	`exists_stronglyContinuousUnitaryGroup` (Bargmann's theorem for $\mathbb{R}$)
Bargmann, the cocycle lemma	`Real.exists_eq_exp_of_localCocycle`, `Real.exists_eq_of_continuousOn_localCocycle`
Theorem 3, DR-PROB ⇒ DR-UNIT	`DrProb.drEv`, then `DrEv.drUnit`
Theorem 3, DR-INFO ⇒ DR-UNIT	`DrInfo.drProb`, then `DrProb.drUnit`
Theorem 3, complete	`schrodinger_tfae` (for $\hbar \neq 0$)
The past clauses of DR-PROB and DR-INFO are needed	`shift_semigroup_counterexample` (from `isEvolution_shiftEvolution`, `not_drEv_shiftEvolution`, `not_drUnit_shiftEvolution`)

Differences from the brief.

- Proposition 1 assumes affinity under all convex combinations, as in finite dimension (`AffineOnDensityOperators`); the proof uses weights other than $\frac{1}{2}$, namely $(1 + |\langle\psi|\phi\rangle|)/2$ in the two-state decomposition and the eigenvalues of the canonical expansion. A hypothesis for equal mixtures only, like the one of the brief "Schrödinger equation is equivalent to information conservation", is enough (`affine_pure_state_correspondence_of_map_mix`), and needs no continuity: density operators have operator norm at most one, so the map is bounded on them, and a bounded map that preserves midpoints preserves all convex combinations. On a segment, subtracting the affine interpolation leaves a bounded g on $[0, 1]$

with $g(0) = g(1) = 0$ and $g(t) = \frac{1}{2}g(2t)$ on $[0, \frac{1}{2}]$, $g(t) = \frac{1}{2}g(2t-1)$ on $[\frac{1}{2}, 1]$, so $\sup \|g\|$ halves and is zero. The same holds in finite dimension. The trace-norm contraction of affine maps (in `TraceNorm`) is not used: mixed states follow from pure states without continuity.

- Proposition 1, (i) $\Rightarrow$ (iii), replaces the finite-dimensional Parseval averaging, which needs the trace of the identity, by a two-dimensional argument. Let $r \perp s$ with images a, b and suppose $\langle a|b\rangle = \delta \in (0,1)$. For the normalized $f_\pm$ of $a \pm b$, the number $z(x) = \langle f_-|T(P_x)f_+\rangle$ lies on the circle $|z|^2 = (1-\delta^2)/4$, and affinity sends pairs at overlap $\frac{1}{2}$ to perpendicular points. The vectors r, $(r+s)/\sqrt{2}$ and $(r+is)/\sqrt{2}$ are pairwise at overlap $\frac{1}{2}$, and three pairwise perpendicular nonzero vectors do not fit in the plane. So orthogonal pure states stay orthogonal, overlaps are preserved, and Wigner's theorem in any dimension with surjectivity gives W. Mixed states: cut the canonical expansion $\rho = \sum_i \mu_i P_{b_i}$ after weight m, $\rho = m\rho_F + (1-m)\sigma_F$; T and conjugation by W agree on the finite mixture ρ_F, so they differ on ρ by at most $2(1-m)$ in operator norm, and $m \to 1$.

- Continuity at 0^+ is in operator norm, which trace-norm continuity implies, and is used only on pure states.

- DR-UNIT is a strongly continuous unitary group, and DR-SCEQ allows an unbounded H. For unbounded H the Schrödinger equation holds only for initial vectors in the domain, and `SolvesSchrodinger` requires that the domain is invariant. DR-SCEQ asks for bounded operators U_t with $U_0 = 1$, not for unitaries; unitarity and the group law follow.

- The "reconstructed past" clauses of DR-PROB and DR-INFO are kept: every pure state has a pure past in DR-PROB, every density operator has a past in DR-INFO. In finite dimension they were dropped, but in infinite dimension they are needed (see below).

- DR-PROB $\Rightarrow$ DR-UNIT goes through DR-EV. Born preservation gives injectivity on pure states (two states with the same image have transition probability 1), and the pure past gives surjectivity; then Proposition 1. In finite dimension the route was Wigner's theorem without surjectivity, which in infinite dimension gives only an isometry.

- The lift (`exists_stronglyContinuousUnitaryGroup`) replaces the finite-dimensional argument through $A_t = \exp(tL)$ on $\mathrm{End}(E)$, which needs norm continuity of the conjugation group. Anti-unitaries are excluded by $\Phi_t = \Phi_{t/2}^2$ as before. The unitaries u_t are then determined up to a phase: an operator commuting with every pure state is a scalar (`exists_eq_smul_of_proj_eq`). Continuity of $t \mapsto P_{u_t\psi}$ at 0^+ extends to every $t \geq 0$, since u_{t_0-h} is $u_h^\dagger u_{t_0}$ up to phase, and by polarization to $t \mapsto |u_t x\rangle\langle u_t y|$. For a fixed unit e, the phase with $\langle e|U_t e\rangle > 0$ makes $U_t x = \langle U_t e|e\rangle^{-1} |u_t x\rangle\langle u_t e|\, e$ strongly continuous near 0. Its multiplier $\omega(a,b) = \langle e|U_a U_b e\rangle \,/\, \langle e|U_{a+b}e\rangle$, with $U_a U_b = \omega(a,b)U_{a+b}$, is continuous, of modulus 1, near 1 and a local 2-cocycle, so $\omega(a,b) = e^{i(f(a)+f(b)-f(a+b))}$ with continuous f. Then $V_t = e^{-if(t)}U_t$ is a local homomorphism on $[0,\varepsilon]$. It extends to $t \geq 0$ by $V_{t/n}^n$ and to $\mathbb{R}$ by $V_{-t} = V_t^\dagger$, and the group law carries strong continuity from 0 to every t.

- The cocycle lemma is proved without a mollifier and without derivatives. Integrate the cocycle identity over $c \in [0,\varepsilon]$ at (a,b,c), and over u between 0 and b at (a,u,ε). With $\Phi(x) = \int_0^\varepsilon \theta(x,c)\,dc$ and $K(x) = \int_0^x \theta(u,\varepsilon)\,du$ this gives $\varepsilon\theta(a,b) = \Phi(a) + \Phi(b) - \Phi(a+b) + K(a+b) - K(a) - K(b)$, so $f = (\Phi - K)/\varepsilon$. The interval may be one-sided, which the lift needs, since U_t is given only for $t \geq 0$.

- DR-UNIT $\Rightarrow$ DR-SCEQ takes $H = -\hbar T$ with T the self-adjoint closure of the candidate generator, $U_t = e^{itT}$ (PhyslibAlpha's Stone theorem). DR-SCEQ $\Rightarrow$ DR-UNIT: with $T = -\hbar^{-1}H$ and $V_t = e^{itT}$ from the spectral theorem, $U_t\psi$ and $V_t\psi$ solve $w' = iTw$ with the same initial value for ψ in the domain, so they agree ($\|w_1 - w_2\|^2$ has derivative zero because T is symmetric); the domain is dense, so $U_t = V_t$.

- As in finite dimension, $\hbar \neq 0$ is assumed in DR-UNIT $\Leftrightarrow$ DR-SCEQ.

Problems found in the brief.

- In infinite dimension the reconstructed past clauses of DR-PROB and DR-INFO cannot be dropped. On $L^2(0,\infty)$ the right shifts $(V_t f)(x) = f(x-t)$ for $x > t$ and 0 otherwise form a strongly continuous semigroup of isometries. $\Phi_t(\rho) = V_t \rho V_t^\dagger$ satisfies `IsEvolution`, maps pure states to pure states, preserves the Born rule and the entropy, but is not a bijection of the pure states and so satisfies neither DR-EV nor DR-UNIT. Formalized as `shift_semigroup_counterexample`. $L^2(0,\infty)$ is the closed subspace of $L^2(\mathbb{R})$ orthogonal to the indicators of the sets of finite measure in $(-\infty, 0)$ (`halfLine`, as the type `HalfLine`), and V_t is translation by $t \geq 0$ (`rightShift`), strongly continuous because translation is continuous on $L^2(\mathbb{R})$. Continuity of $t \mapsto V_t \rho V_t^\dagger$ at 0^+ in operator norm uses that density operators are trace class, hence compact: $\|V_t \rho V_t^\dagger - \rho\| \leq 2\|(V_t - 1)\rho\|$, and bounded operators that tend to 0 pointwise tend to 0 uniformly on the compact closure of the image of the unit ball (`tendsto_mul_of_isCompactOperator`). For $t > 0$ the normalized indicator e of $(0,t)$ is orthogonal to the range of V_t, so $|e\rangle\langle e|$ is Φ_t of no pure state (`not_bijOnPure_conj`). The brief's abstract claims the infinite-dimensional case, and its definitions do contain past states, so the theorem holds as stated once the past is read as part of DR-PROB and DR-INFO.

- "The phases can be chosen so that $U_0 = I$ and $U_{t+s} = U_t U_s$" is Bargmann's lifting theorem for $\mathbb{R}$; the brief neither proves nor cites it. In infinite dimension the finite-dimensional trick is not available, because the conjugation group is only strongly continuous. The proof above is Bargmann's 1954 idea specialized to $\mathbb{R}$, with continuous instead of measurable multipliers.

- Proposition 1 needs no continuity in any dimension, so the continuity in the brief's hypotheses is redundant there. The Remark's weakening to injectivity does not extend to infinite dimension: conjugation by the shift on $\ell^2(\mathbb{N})$ is affine and injective on pure states but not onto them (`shift_preserves_not_unitary`).

- For unbounded H, DR-SCEQ's "$\psi(t) = U(t)\psi(0)$ solves the Schrödinger equation" must be restricted to $\psi(0)$ in the domain of H, which the brief leaves implicit. Otherwise no problems beyond those of the finite-dimensional part.